Name: Cao kaihan Student ID: 23097881D

Question 1

(a) we use Dijkstra's algorithm to find the minimum time.

Iter.1: $D(1,2) = 58$ $D(1,3) = 18$ $D(1,4) = 50$

$D(1,5) = D(1,6) = D(1,7) = D(1,8) = \infty$

we choose 3 for the next expansion.

Iter.2 we meet: 5, 6, 7.

$D(1,2) = 58$ $D(1,4) = 50$ $D(1,6) = 18 + 21 = 39$

$D(1,5) = 18 + 72 = 90$ $D(1,7) = 18 + 35 = 53$ $D(1,8) = \infty$

we choose 6 for the next expansion.

Iter.3 we meet 8, 7

$D(1,2) = 58$ $D(1,4) = 50$ $D(1,5) = 90$

$D(1,7) = \min\{53, 39+13\} = 52$ $D(1,8) = 58+39 = 97$

we choose 4 for the next expansion.

Iter.4 we meet 5.

$D(1,2) = 58$ $D(1,5) = \min\{90, 50+62\} = 90$

$D(1,7) = 52$ $D(1,8) = 97$

we choose 7 for the next expansion.

Iter.5 we meet 8.

$D(1,2) = 58$ $D(1,5) = 90$ $D(1,8) = \min\{97, 52+38\} = 90$

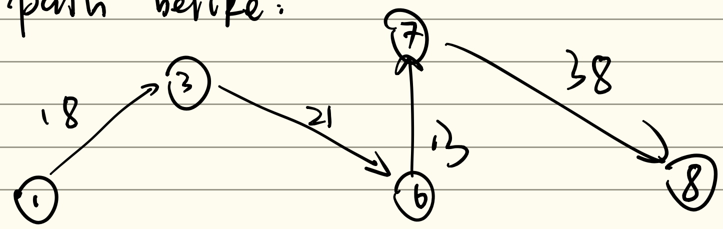
we choose 2 for next expansion.

Iter.6 No unvisited nodes

$D(1,5) = 90$ $D(1,8) = 90$

5, 8 expand in this iter. and the next iter., respectively.

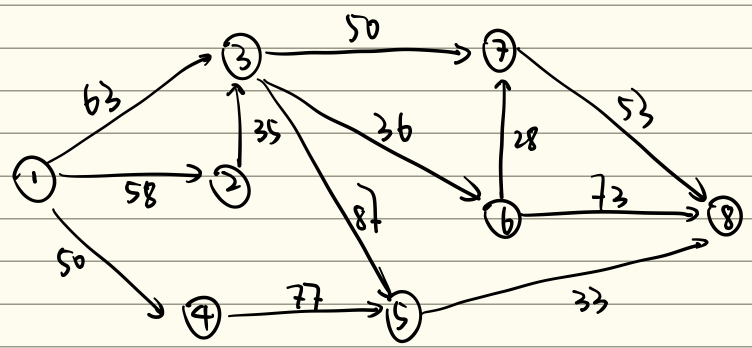
So the path be like:



$\text{Time} = 18 + 21 + 13 + 38 = 90$ mins

(b)

The new weighted graph:



Also use Dijkstra's algorithm.

Iter. 1 $D(1,2) = 58$ $D(1,3) = 63$ $D(1,4) = 50$

$D(1,5) = D(1,6) = D(1,7) = D(1,8) = \infty$

we choose ④ for the next expansion

Iter. 2 we meet ⑤.

$D(1,2) = 58$ $D(1,3) = 63$ $D(1,5) = 50 + 77 = 127$

$D(1,6) = D(1,7) = D(1,8) = \infty$

we choose ② for the next expansion.

Iter. 3 we meet ③

$D(1,3) = \min\{63, 58 + 35\} = 63$ $D(1,5) = 127$

$D(1,6) = D(1,7) = D(1,8) = \infty$

choose ③ for the next expansion.

Iter. 4 we meet ⑤, ⑥, ⑦.

$D(1,5) = \min\{127, 63+87\} = 127$ $D(1,6) = 63 + 36 = 99$

$D(1,7) = 63 + 50 = 113$ $D(1,8) = \infty$

choose ⑥ for the next expansion.

Iter. 5 we meet ⑦, ⑧

$D(1,5) = 127$ $D(1,7) = \min\{113, 99+28\} = 113$ $D(1,8) = 99 + 73 = 172$

choose ⑦ for the next expansion.

Iter. 6 we meet ⑧

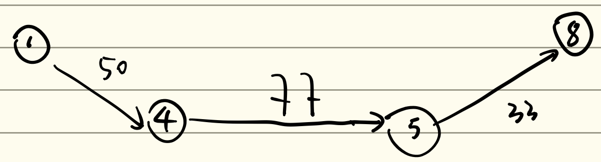
$D(1,5) = 127$ $D(1,8) = \min\{172, 113 + 53\} = 166$

choose ⑤ for the next expansion.

Iter. 7 we meet ⑧

$D(1,8) = \min\{166, 127 + 33\} = 160$ - Finish.

∴ The shortest path be like



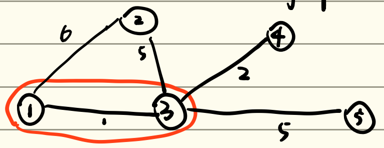
Time = 50 + 77 + 33

= 160 mins

Question 2

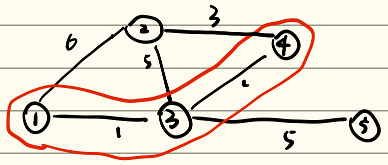
Start from ①. Iter. 1 consider edge: $D(1,3)=1$, $D(1,2)=6$

Take ③ into the graph.



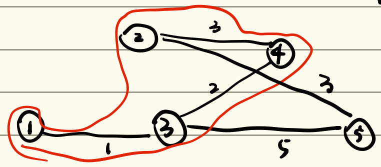
Iter. 2 consider edge: $D(1,2)=6$ $D(3,2)=5$ $D(3,4)=2$ $D(3,5)=5$

Take ④ into the graph.



Iter. 3 consider edge $D(1,2)=6$ $D(3,2)=5$ $D(4,2)=3$ $D(3,5)=5$

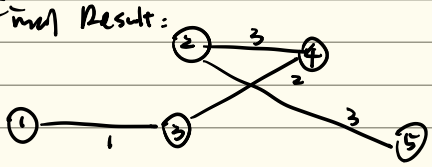
Take ② into the graph.



Iter. 4 consider edge $D(2,5)=3$ $D(3,5)=5$

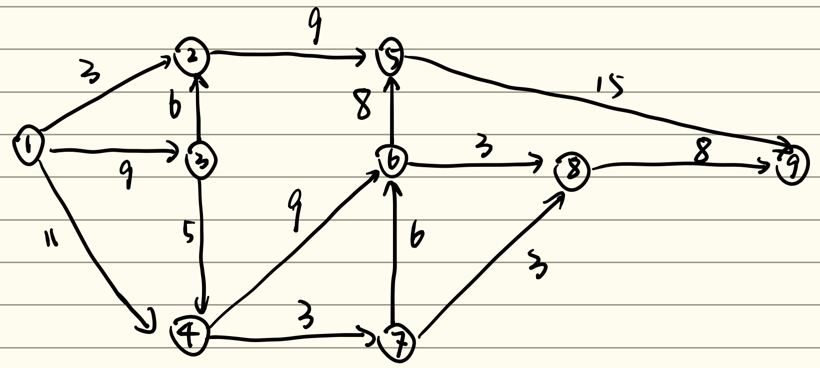
Take ⑤ into the graph. - Finish

Final Result:



The minimum weight $= 1 + 2 + 3 + 3 = 9$

Question 3

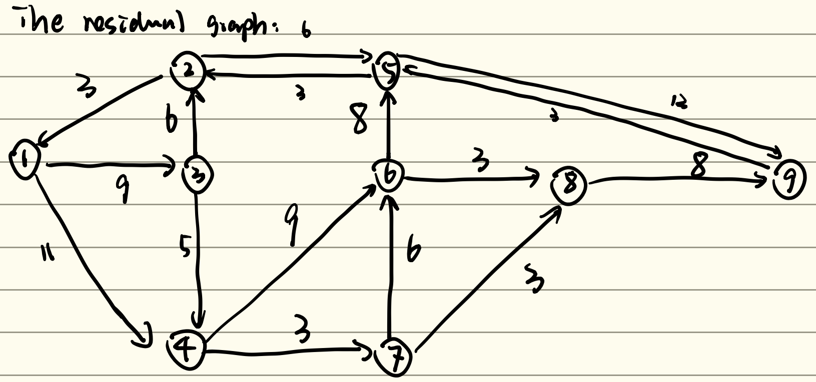


(a) The augumenting path with 3 edges:

$① \longrightarrow ② \longrightarrow ⑤ \longrightarrow ⑨$

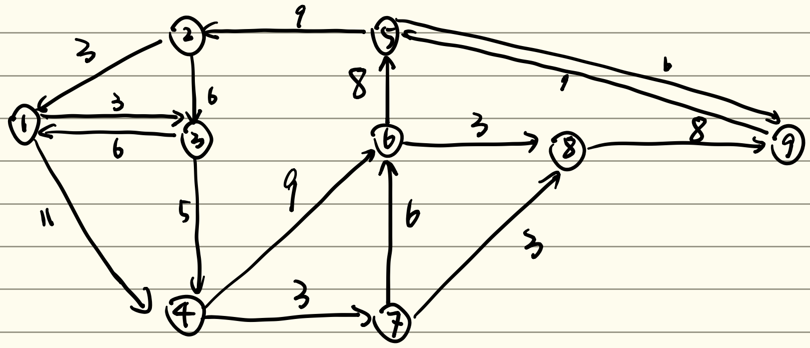
$R(x) = 3$

The residual graph:

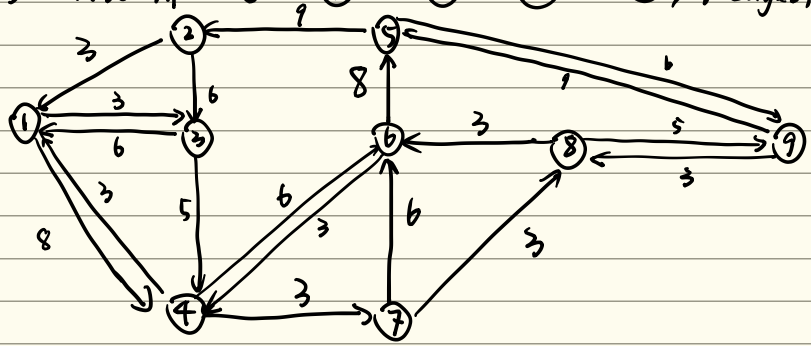


(b) EK algorithm: Each iteration, find the augumenting path with the smallest number of edges.

Iter.2 choose AP $① \to ③ \to ② \to ⑤ \to ⑨$, 4 edges. $R(x) = 6$

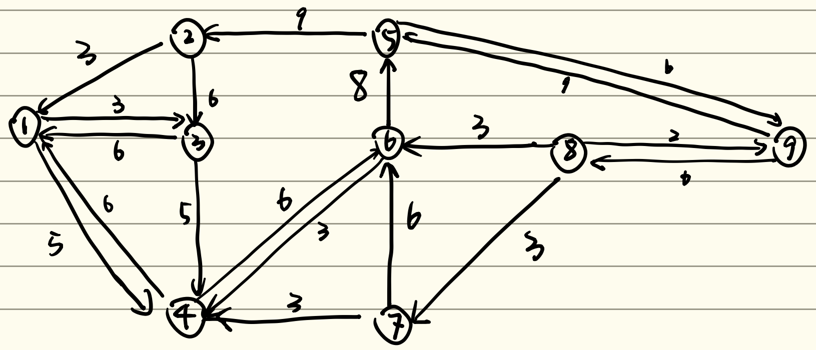


Iter.3 choose AP $① \to ④ \to ⑥ \to ⑧ \to ⑨$, 4 edges, $R(x) = 3$

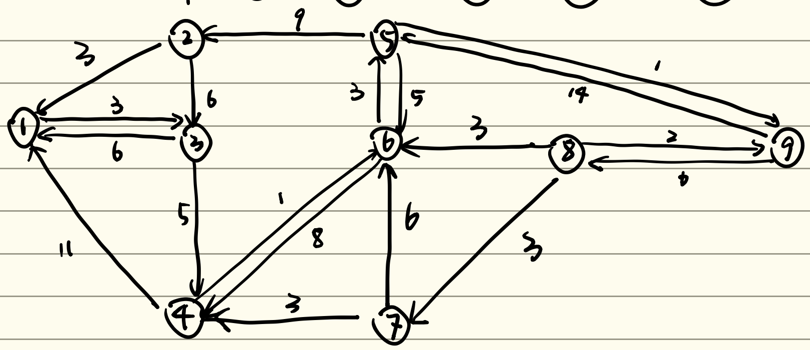


3(b), cond.

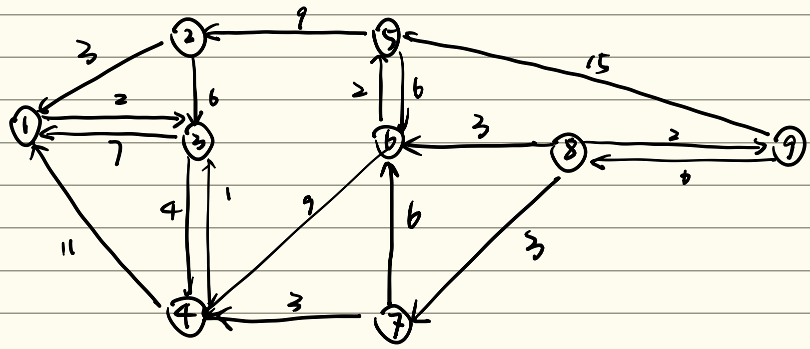
Iter.4 choose AP ① → ④ → ⑦ → ⑧ → ⑨ . 4 edges. $R(x) = 3$



Iter.5 choose AP ① → ④ → ⑥ → ⑤ → ⑨ 4 edges, $R(x) = 5$

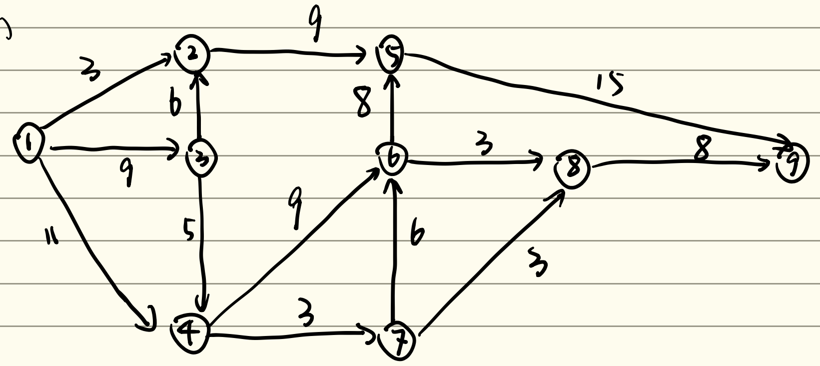


Iter.6 choose AP ① → ③ → ④ → ⑥ → ⑤ → ⑨ , 5 edges, $R(x) = 1$



Finish. no AP anymore. The final maximum flow value $= 1 + 5 + 3 + 3 + 6 + 3 = 21$

3 (c)



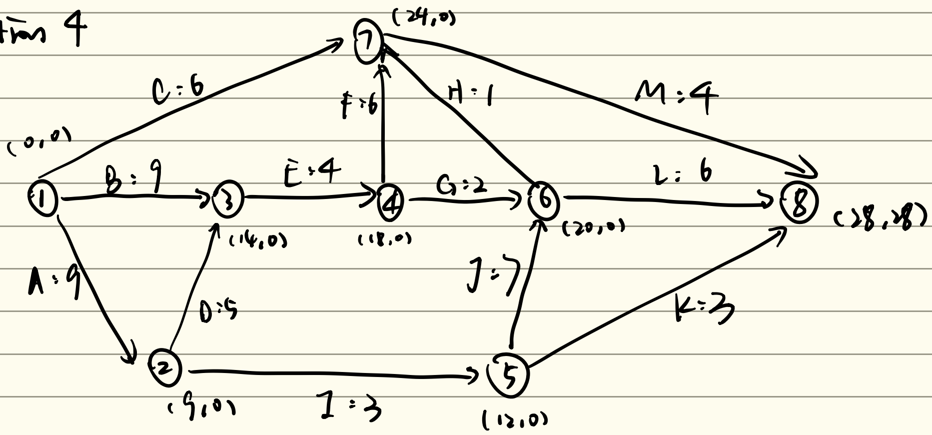
From (b), the nodes that can be achieved from the residual graph: {1, 3, 4}.

$\therefore I = \{1, 3, 4\}, \quad O = \{2, 5, 6, 7, 8, 9\}$

$$Cut_{(min)} = D(1,2) + D(4,6) + D(4,7) + D(3,2)$$
$$= 3 + 9 + 3 + 6 = 21$$

$\therefore$ the minimum cut $= 21$ equals to max flow.

Question 4



(a) $ET(1) = 0$ $\quad ET(2) = ET(1) + D(1,2) = 0 + 9 = 9$ $\quad ET(3) = \max\{ET(1) + D(1,3), ET(2) + D(2,3)\}$
$= \max\{9, 9+5\} = 14$

$ET(4) = ET(3) + D(3,4) = 14 + 4 = 18$ $\quad ET(5) = ET(2) + D(2,5) = 9 + 3 = 12$

$ET(6) = \max\{ET(4) + D(4,6), ET(5) + D(5,6)\}$
$= 20$

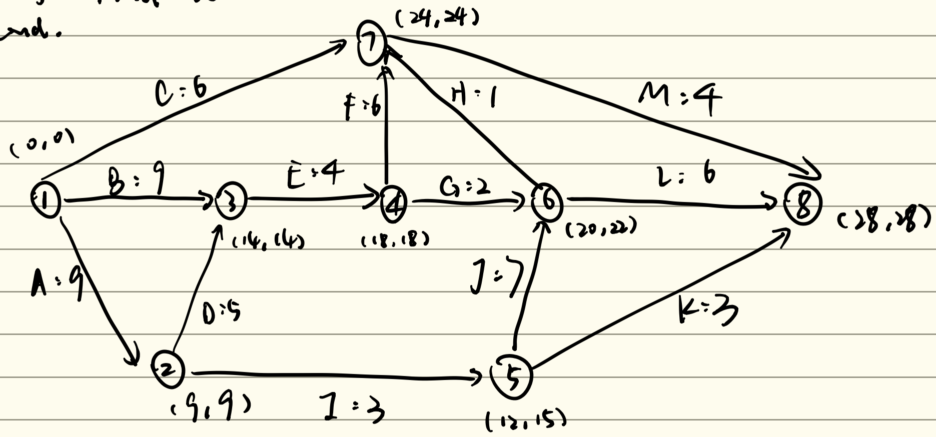
$ET(7) = \max\{ET(1) + D(1,7), ET(4) + D(4,7), ET(6) + D(6,7)\} = 24$

$ET(8) = \max\{ET(5) + D(5,8), ET(6) + D(6,8), ET(7) + D(7,8)\} = 28$

Summary:

$$\begin{cases} ET(1) = 0, \ ET(2) = 9 \quad ET(3) = 14 \\ ET(4) = 18, \ ET(5) = 12 \quad ET(6) = 20 \\ ET(7) = 24 \quad ET(8) = 28. \end{cases}$$

4(a) Find the LT.
cont.



$LT(8) = ET(8) = 28$

$LT(7) = LT(8) - D(7,8) = 24$

$LT(6) = \min\{LT(8) - D(6,8), LT(7) - D(7,6)\}$
$= 22$

$LT(5) = \min\{LT(8) - D(8,5), LT(6) - D(6,5)\}$
$= 15$

$LT(4) = \min\{LT(6) - D(4,6), LT(7) - D(4,7)\}$
$= 18$

$LT(3) = LT(4) - D(3,4) = 14$

$LT(2) = \min\{LT(3) - D(2,3), LT(5) - D(2,5)\}$
$= 9$

$LT(1) = 0$

(b) Find the total float and free float (for activities)

A: $TF(1,2) = LT(2) - ET(1) - D(1,2)$
$= 9 - 0 - 9 = 0$

$FF(1,2) = ET(2) - ET(1) - D(1,2)$
$= 9 - 0 - 9 = 0$

B: $TF(1,3) = LT(3) - ET(1) - D(1,3)$
$= 14 - 0 - 9 = 5$

$FF(1,3) = ET(3) - ET(1) - D(1,3)$
$= 14 - 0 - 9 = 5$

C: $TF(1,7) = LT(7) - ET(1) - D(1,7)$
$= 24 - 6 - 0 = 18$

$FF(1,7) = ET(7) - ET(1) - D(1,7)$
$= 24 - 0 - 6 = 18$

D: $TF(2,3) = LT(3) - ET(2) - D(2,3)$
$= 14 - 9 - 5 = 0$

$FF(2,3) = ET(3) - ET(2) - D(2,3)$
$= 14 - 9 - 5 = 0$

E: $TF(3,4) = LT(4) - ET(3) - D(3,4)$
$= 18 - 14 - 4 = 0$

$FF(3,4) = ET(4) - ET(3) - D(3,4)$
$= 18 - 14 - 4 = 0$

F: $TF(4,7) = LT(7) - ET(4) - D(4,7)$
$= 24 - 18 - 6 = 0$

$FF(4,7) = ET(7) - ET(4) - D(4,7)$
$= 24 - 18 - 6 = 0$

G: $TF(4,6) = LT(6) - ET(4) - D(4,6)$
$= 22 - 18 - 2 = 2$

$FF(4,6) = ET(6) - ET(4) - D(4,6)$
$= 20 - 18 - 2 = 0$

H: $TF(6,7) = LT(7) - ET(6) - D(6,7)$
$= 24 - 20 - 1 = 3$

$FF(6,7) = ET(7) - ET(6) - D(6,7)$
$= 24 - 20 - 1 = 3$

I: $TF(2,5) = LT(5) - ET(2) - D(2,5)$
$= 15 - 9 - 3 = 3$

$FF(2,5) = ET(5) - ET(2) - D(2,5)$
$= 12 - 9 - 3 = 0$

J: $TF(5,6) = LT(6) - ET(5) - D(5,6)$
$= 22 - 12 - 7 = 3$

$FF(5,6) = ET(6) - ET(5) - D(5,6)$
$= 20 - 12 - 7 = 1$

K: $TF(5,8) = LT(8) - ET(5) - D(5,8)$
$= 28 - 12 - 3 = 13$

$FF(5,8) = ET(8) - ET(5) - D(5,8)$
$= 28 - 12 - 3 = 13$

L: $TF(6,8) = LT(8) - ET(6) - D(6,8) = 28 - 20 - 6$
$= 2$

$FF(6,8) = ET(8) - ET(6) - D(6,8)$
$= 28 - 20 - 6 = 2$

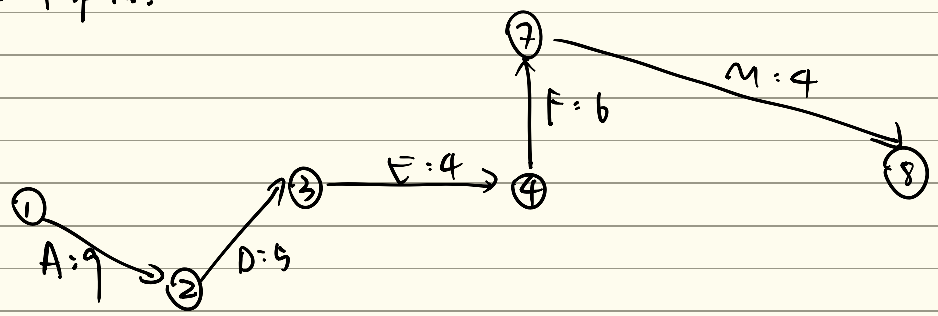
M: $TF(7,8) = LT(8) - ET(7) - D(7,8)$
$= 28 - 24 - 4 = 0$

$FF(7,8) = ET(8) - ET(7) - D(7,8)$
$= 0$

4(c). Find the critical path:

Link all activity with TF = 0, they're: A, D, E, F, M

Critical path:



Duration = 9 + 5 + 4 + 6 + 4 = 28 Days